

Product Requirements Document (PRD)

OpenScholar

Version: 1.0

Status: MVP Planning (Updated)

Table of Contents

- [1. Product Overview](#)
 - [2. Goals and Success Metrics](#)
 - [3. Target Users](#)
 - [4. Feature Prioritization \(MoSCoW\)](#)
 - [5. Differentiation](#)
 - [6. User Flow](#)
 - [7. System Architecture](#)
 - [8. Non-Functional Requirements](#)
 - [9. Constraints](#)
 - [10. Risks and Mitigation](#)
 - [11. Roadmap](#)
 - [12. MVP Definition](#)
 - [13. Acceptance Criteria](#)
 - [14. Product Philosophy](#)
-

1. Product Overview

1.1 Vision

To build a global open-access academic research platform where students can publish, discover, and access research without institutional or financial barriers.

1.2 Mission

- Democratize access to academic knowledge
 - Enable student-first research publishing
 - Build a scalable academic knowledge network
 - Replace outdated research platforms with modern systems
-

1.3 Problem Statement

- Research is restricted by paywalls
 - Platforms require institutional email access
 - Student research lacks visibility
 - Existing platforms have poor usability
 - No centralized academic repository in Bangladesh
-

1.4 Solution Summary

OpenScholar provides:

- Open-access research publishing
 - Free reading and downloading
 - Simple publishing workflow
 - Modern UI/UX
 - Scalable architecture
-

2. Goals and Success Metrics

2.1 Business Goals

- Launch MVP within 4–5 months
 - Onboard initial university
 - Expand to multiple universities within 1 year
-

2.2 Product Goals

- Paper publishing within 5 minutes
 - Search response time under 2 seconds
 - High readability experience
 - Free access to all content
-

2.3 Success Metrics

Metric	Target
Monthly Active Users	10,000+
Papers Published	5,000+

Metric	Target
Average Session Time	5+ minutes
Upload Success Rate	99%
Search Response Time	< 2 seconds
Retention Rate	40%+

3. Target Users

Primary Users

- Undergraduate students
- Graduate students
- Research scholars

Secondary Users

- Faculty members
- Universities
- General researchers

4. Feature Prioritization (MoSCoW)

4.1 Must Have (MVP Core)

Authentication

- Email/password login
- Email verification
- Password reset
- Role-based access control

User Profile

- Basic profile
- Publication list

Research Publishing

- PDF upload ($\leq 50\text{MB}$)
- Metadata input:

- Title
- Abstract
- Keywords
- University and department selection
- Draft saving
- Version control (mandatory for edits)
- Multi-author support
- Author ordering (1st, 2nd, etc.)
- Submission status management

Admin Moderation

- Approve/reject submissions
- Publish approved papers
- Moderation logging

Paper Viewing

- Paper details page
- PDF viewer
- Download functionality

Search and Discovery

- Keyword search
- Filters:
 - University
 - Department
 - Year

User Engagement

- Reactions (1 per user per paper)
- Comments
- Reply to comments (1-level nesting)
- Share (link/social)

Basic Analytics

- View tracking
- Download tracking

Core Non-Functional Requirements

- Page load under 3 seconds
- Search response under 2 seconds
- Secure authentication
- HTTPS communication
- 99% uptime

4.2 Should Have

- Dashboard (user analytics, most viewed papers)
 - Citation export
 - Multilingual support (English & Bangla)
 - Enhanced search filters (category)
 - Security enhancements
-

4.3 Could Have

- Focus reading mode
 - Bookmark/save papers
 - Notifications
 - Advanced analytics (author-level, ranking)
 - Caching and CDN
-

4.4 Won't Have (MVP Excluded)

- DOI integration
 - Peer review system
 - AI recommendation engine
 - Automated plagiarism detection
 - Paid subscription system
-

5. Differentiation

- Clean, user-focused interface
 - Simplified publishing workflow
 - Open-access model
 - Student-centered design
 - Engagement-driven academic platform
-

6. User Flow

Publishing Flow

```
graph TD;
    A[Register] --> B[Login];
    B --> C[Submit Paper];
    C --> D[Admin Review];
    D --> E[Publish];
```

Discovery Flow

```
graph TD;
    A[Search] --> B[Filter];
    B --> C[Open Paper];
    C --> D[Read];
    D --> E[Download];
    E --> F[Interact (react/comment/share)];
```

User Flow Diagram

```
flowchart LR
    A[Register] --> B[Login]
    B --> C[Submit Paper]
    C --> D[Admin Review]
    D -->|Approved| E[Publish]
    D -->|Rejected| F[Revision Required]
    E --> G[Readers Discover Paper]
    G --> H[Read & Download]
    H --> I[React / Comment / Share]
```

7. System Architecture

Layer	Technology
Frontend	Next.js
Backend	API-based architecture
Database	PostgreSQL
Storage	Cloud storage (S3 or equivalent)
Authentication	JWT/session-based

High-Level Architecture Diagram

```
flowchart TB
    U[Users] --> F[Next.js Frontend]
    F --> A[Backend API]
    A --> DB[(PostgreSQL)]
    A --> S3[(Cloud Storage)]
    A --> AUTH[Authentication Service]
```

8. Non-Functional Requirements

Must

- Performance: <3 seconds load time
- Security: encryption and HTTPS
- Reliability: 99% uptime

Should

- Scalability: multi-university support
- Maintainability: modular architecture

Could

- Zero-downtime deployment
- Advanced monitoring

9. Constraints

- Limited moderation resources
 - Initial deployment limited to one university
 - No DOI integration in MVP
 - No automated plagiarism detection
-

10. Risks and Mitigation

Risk	Mitigation
Fake submissions	Admin moderation
Plagiarism	Manual review
Low adoption	University onboarding
System overload	Cloud scaling

11. Roadmap

Phase 1 (MVP)

- Authentication
- Publishing system
- Search
- Admin panel
- Engagement features

Phase 2

- Analytics improvements
- UX improvements
- Search optimization

Phase 3

- AI features
- Recommendation system

Phase 4

- Multi-university expansion
- Global scaling

Roadmap Visualization

```
gantt
    title OpenScholar Roadmap
    dateFormat YYYY-MM-DD

    section MVP
    Authentication      :done, a1, 2026-01-01, 20d
    Publishing System   :a2, after a1, 40d
    Search System       :a3, after a2, 20d

    section Growth
    Analytics           :b1, after a3, 20d
    UX Improvements     :b2, after b1, 20d

    section Scale
    AI Features         :c1, after b2, 40d
    Multi-University    :c2, after c1, 60d
```

12. MVP Definition

Includes

- Authentication
- Profile management
- Paper upload (multi-author)
- Admin approval
- Search
- Reading & download
- Reactions, comments, sharing

13. Acceptance Criteria

Paper Upload

- PDF $\leq$ 50MB

- Metadata required
- Supports multiple authors with order
- Status = pending

Search

- Results within 2 seconds
- Filters operational

Authentication

- Email verification required
- Invalid login blocked

Engagement

- One reaction per user per paper
 - Comment and reply functional
-

14. Product Philosophy

- Open access
- Simplicity
- Clean design
- Accessibility
- Scalability